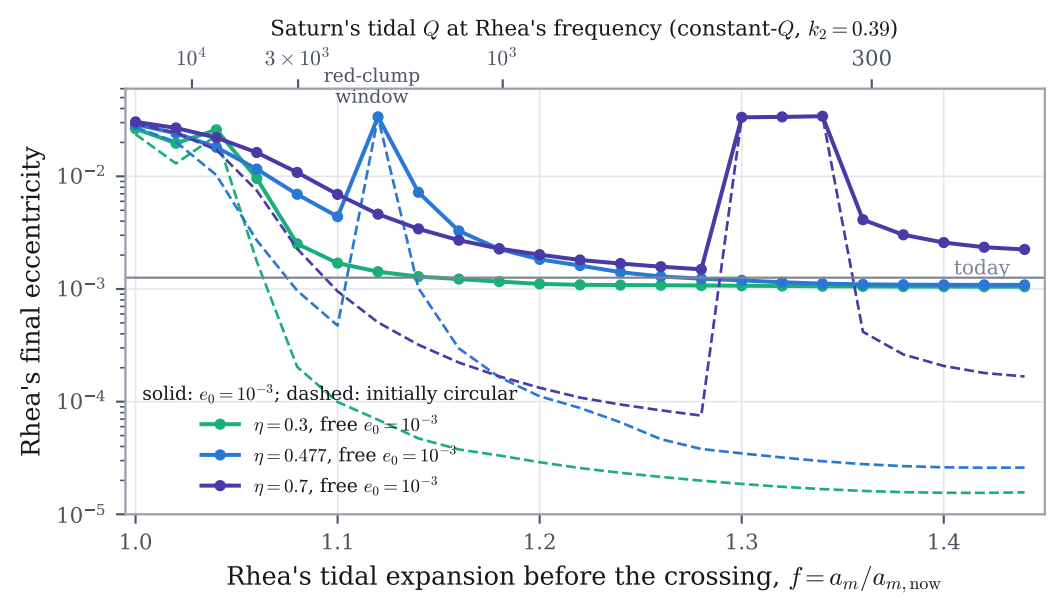
Saturn's tidal Q at Rhea's frequency (constant-Q, $k_2 = 0.39$)
10^4
3×10^3
10^3
300
red-clump window
Rhea's final eccentricity
10^{-2}
10^{-3}
10^{-4}
10^{-5}
today
solid: $e_0 = 10^{-3}$; dashed: initially circular
$\eta = 0.3$, free $e_0 = 10^{-3}$
$\eta = 0.477$, free $e_0 = 10^{-3}$
$\eta = 0.7$, free $e_0 = 10^{-3}$
1.0
1.1
1.2
1.3
1.4
Rhea's tidal expansion before the crossing, $f = a_m / a_{m,\,\mathrm{now}}$